AeroMesh: FPGA-Accelerated LoRa Mesh Network for Real-Time Drone Collision Avoidance

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Background:

As the number of drones operating in the same airspace increases, so does the risk of mid-air collisions. Relying on individual communication channels or a centralized controller to coordinate each drone can become inefficient as the number of drones increases. This project aims to establish a distributed LoRa mesh network that can enable drones to exchange real-time information, including position, velocity, and intended trajectory. This allows drones to detect potential collisions and autonomously adjust their trajectories appropriately.

Project Needs:

1. Communication: Reliable wireless communication between drone nodes and the ability for drones to share a common LoRa communication channel to exchange compact, real-time drone state information; packet error detection and handling.
2. Multi-Hop Networking: Drones must be able to forward packets to other drones, and they must still be able to communicate when a direct link between two drones is unavailable; dynamic route selection and recovery from a lost link.
3. Collision Detection/Avoidance: Obtain each drone's position, velocity, heading, and timestamp, and predict/calculate future drone positions to determine collision risk; Provide sufficient warning before potential collision.
4. System Monitoring: Display network status, drone positions/trajectories, collision warning, time-to-collision, and log system performance.

Project Scope:

1. Hardware: Effectively utilize existing, programmable drone platforms for flight control and employ an FPGA to accelerate time-critical packet-processing and collision-prediction operations to reduce processing latency.
2. Networking: Establish a LoRa-based wireless communication using multi-hop communication and a basic multi-hop network architecture to send small telemetry packets.
3. Collision Avoidance: Position/velocity-based trajectory prediction, time-to-collision calculation, integration with an existing flight controller, autonomous recovery to the original trajectory.
4. Testing: Demonstrate 1-hop communication, 2-3-hop communication, direct-link failure, route recovery, collision prediction testing, and autonomous avoidance; compare FPGA-accelerated and software-based implementations to evaluate processing latency, system response time, and FPGA resource utilization.

Potential Technical Considerations:

1. LoRa Communication: communication range, data rate, packet size, shared-channel access, packet loss, interference, and communication latency.
2. Mesh Networking: Neighbor discovery, dynamic routing, link quality, network congestion, and recovery from lost connections.
3. Collision Detection/Avoidance: position and velocity accuracy, trajectory prediction, time-to-collision, safety distance, communication delay, and selection of an appropriate evasive maneuver.
4. FPGA / Drone Integration: FPGA acceleration of time-critical packet processing and collision calculations; FPGA resource utilization and processing latency, comparison with software-based processing, flight controller communication, autonomous flight commands, and manual/failsafe controls.

Deliverables:

1. Prototype: A working prototype of the mesh network that showcases dynamic rerouting for communication and autonomous recovery from a drone based on collision prediction.
2. Technical Documentation: Documentation that outlines the network and hardware design, implementation, and methodology.
3. Project Report: A formal report describing results, limitations, and future work.
4. Presentation: A presentation demonstrating the project and its proficiency to ORNL and university staff, students, and families.

Assumptions and Dependencies:

1. Guidance and collaboration with ORNL research experts as an ORNL-sponsored project.
2. Necessary resources for drone integration and network deployment for testing.